

HACKING WIRELESS NETWORKS LAB REPORT

BACKGROUND	
Course:	CIS 170 - Hands-On Ethical Hacking and Network Defense
Term:	Spring 2026
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Submission Platform:	Canvas
Report Date:	2026-02-18
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Lab Title:	Hacking Wireless Networks
Module Reference:	Hacking Wireless Networks
Lab Type:	Live Virtual Machine Lab
Date Completed:	2026-02-18
Time Spent:	Approximately 1 hour
ENVIRONMENT	
Platforms and Systems:	
- Primary environment: Practice Labs / Live Virtual Machine Labs controlled sandbox. - Primary systems available in the topology: PLABDC01 (Windows Server 2019 - Domain Controller), PLABDM01 (Windows Server 2019 - Domain Member), PLABWIN10 (Windows 10 - Domain Member), and PLABKALI01 (Kali 2019.2 - Linux Kali Workstation). - The work centered mainly on PLABWIN10 using Internet Explorer, NetSurveyor, and the netsh command for wireless discovery, protocol analysis, and threat identification. - Remote access to the virtual machines was provided through the course platform.	
Context and Constraints:	
- The module focused on wireless protocols (WEP, WPA, WPA2), revealing hidden SSIDs, wireless threats, hacking methodology, and hardening techniques. - Practice Labs flagged the module as legacy content using deprecated software, so behavior could vary and some steps were best interpreted as guided sandbox demonstrations. - Windows Updates were disabled in the Practice Labs environment, so software behavior reflected a controlled legacy setup rather than a current hardened enterprise baseline. - The module aligned most closely with exam objectives 3.2 (Information Security Attack Detection) and 3.3 (Information Security Attack Prevention).	
Authorization and Scope:	
- All documented activity took place only inside approved academic systems and isolated practice-lab sandboxes. - No public IP space, real organizations, or unauthorized targets were involved. - No production infrastructure, third-party systems, or live services were contacted. - The actions documented here were performed for defensive understanding, wireless network attack simulation, and ethical hacking education within the course environment. - Restricted material, live credentials, and personal data are omitted or redacted from this report.	
Work Objectives:	
- Review wireless protocols (WEP, WPA, WPA2), hidden SSID discovery, common wireless threats, and hacking methodology. - Use NetSurveyor and netsh wlan show networks mode=bssid on PLABWIN10 to discover networks and reveal hidden SSIDs. - Identify and document wireless threats (rogue APs, ad-hoc networks, DoS, MAC spoofing, MITM). - Review hardening techniques (encryption, SSID hiding, MAC filtering, least-privilege configuration).	
Hacking Wireless Networks Concepts and Defensive Framing:	
- The lab identified four core phases of wireless attacks: protocol analysis, discovery (hidden SSIDs), threat identification, and hardening. - Attack types reviewed included WEP/WPA weaknesses, hidden SSID revelation, rogue APs, ad-hoc networks, DoS, MAC spoofing, MITM, and de-authentication attacks. - Tools demonstrated both GUI (NetSurveyor) and CLI (netsh) approaches, showing how weak configurations expose wireless networks. - Prevention methods reviewed included strong encryption (WPA2/WPA3), SSID hiding, MAC filtering, regular auditing, and least-privilege configuration, reinforcing that wireless defense requires both technical controls and ongoing monitoring.	

TECHNICAL ANALYSIS

Wireless Discovery and Hidden SSID Revelation on PLABWIN10:

- On PLABWIN10, NetSurveyor was launched from the Intranet Hacking Tools page to perform a full wireless scan.
- The tool displayed a detailed list of all nearby networks, showing SSIDs, channels, signal strength, encryption types (WEP/WPA/WPA2), MAC addresses of access points, and whether SSIDs were hidden or broadcasting.
- Several networks with hidden SSIDs were identified that did not appear in normal Windows Wi-Fi lists.
- The netsh command was executed in an elevated Command Prompt: `netsh wlan show networks mode=bssid`.
- This command successfully revealed the hidden SSIDs, BSSID (MAC addresses), signal strength, and authentication/encryption details for networks that were not broadcasting their names.

Wireless Protocols, Threats, and Hardening Review:

- Wireless protocols were analyzed in detail: WEP (weak RC4 encryption, 24-bit IV, easily cracked), WPA (TKIP, still vulnerable), and WPA2 (AES-CCMP, current standard) were compared side-by-side.
- Common wireless threats were identified and documented: rogue access points, ad-hoc networks, DoS attacks, configuration issues (default admin passwords/SSIDs), passive capturing, MAC spoofing, man-in-the-middle, and de-authentication attacks.
- Hardening techniques were reviewed: enabling strong encryption (WPA2/WPA3 with AES), disabling SSID broadcasting, changing default admin passwords, implementing MAC filtering, reducing radio transmission power, installing antivirus/firewalls, and performing regular network auditing.

KEY COMMANDS, TOOLS, AND ARTIFACTS BY SYSTEM

PLABWIN10 — Wireless Discovery Commands:

netsh wlan show networks mode=bssid

(revealed hidden SSIDs, BSSID, signal strength, authentication/encryption types)

NetSurveyor (GUI wireless scanner launched from Intranet Hacking Tools)

(displayed full list of networks including hidden SSIDs, channels, signal strength, encryption)

Primary Tools and Artifacts Observed:

- NetSurveyor scan results showing visible and hidden SSIDs, channels, signal strength, encryption types (WEP/WPA/WPA2), and MAC addresses of access points.
- netsh wlan show networks mode=bssid output successfully revealing hidden SSIDs that were not broadcasting.
- Detailed protocol comparison (WEP weak RC4, WPA TKIP, WPA2 AES-CCMP) and documented threats (rogue APs, ad-hoc networks, DoS, MAC spoofing, MITM, de-authentication).
- Hardening recommendations including WPA2 encryption, SSID hiding, MAC filtering, and regular auditing.

DEFENSIVE MAPPING AND SUMMARY

Technique Summary:

- The completed work covered wireless protocols (WEP, WPA, WPA2), hidden SSID discovery with NetSurveyor and netsh, common wireless threats, hacking methodology, and practical hardening techniques.
- The recurring theme across the module was that wireless networks are inherently exposed by design and require layered defenses beyond default configurations.

MITRE ATT&CK Mapping:

- Initial Access: T1190 Exploit Public-Facing Application / T1558 Steal or Forge Kerberos Tickets - rogue APs and de-authentication attacks.
- Discovery: T1595.002 Vulnerability Scanning / T1046 Network Service Discovery - NetSurveyor and netsh for wireless network enumeration.
- Credential Access: T1555 Credentials from Password Stores - passive capturing and MITM.
- Impact: T1499 Endpoint Denial of Service - de-authentication and jamming attacks.

Detection Opportunities:

- Unexpected rogue APs, de-authentication frames, MAC address changes, and unusual traffic patterns are strong indicators of attack.
- Wireless IDS/IPS can detect hidden SSID probes, de-auth floods, and unauthorized MAC addresses.
- Regular NetSurveyor or netsh scans combined with centralized logging help identify anomalies.

Defensive Controls:

- Use strong encryption (WPA2/WPA3 with AES), disable SSID broadcasting, and implement MAC filtering.
- Change default admin passwords, reduce transmission power, and enable network auditing.
- Deploy wireless IDS/IPS, regularly scan for rogue APs, and enforce antivirus/firewalls on all clients.
- Apply least-privilege principles and perform regular security assessments of wireless infrastructure.

Completion Verification:

- Work was completed across wireless protocol analysis, hidden SSID discovery with NetSurveyor and netsh, threat identification, and hardening techniques.
- Completion was evidenced by NetSurveyor scans, netsh output revealing hidden SSIDs, and documented threats and controls.
- The actual commands and tool outputs used in the lab are preserved directly in this report for accurate documentation and reproducibility.

Key Findings:

- 1) Wireless networks are inherently exposed because radio signals can be intercepted from outside physical perimeters.
- 2) Hidden SSIDs provide false security - tools like NetSurveyor and netsh easily reveal them.
- 3) WEP and weak WPA configurations are easily cracked; WPA2/WPA3 with AES is the current minimum.
- 4) Common threats (rogue APs, de-auth, MAC spoofing) require layered defenses beyond encryption.
- 5) The strongest defensive takeaway was that regular auditing, strong encryption, and least-privilege configuration are essential to secure wireless networks.

Issues / Deviations / Notes:

- The lab used deprecated software and a legacy environment, so exact tool behavior and scan results could vary slightly.
- All activities were performed only inside the isolated Practice Labs sandbox.

Reflection:

- This lab tied together wireless protocol analysis, discovery techniques, threat identification, hacking methodology, and practical hardening into a complete attack-defense workflow.
- The strongest takeaway was that wireless networks are dangerous because they are publicly exposed by design, making strong encryption, regular auditing, and layered controls essential for defense.